

Mathematics I

CSIT — 1st Semester — 2080 — Answer Sheet
Subject Code: MTH117

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. (a) If $\vec{a} = (4, 0, 3)$ and $\vec{b} = (-2, 1, 5)$, find $|\vec{a}|$, $3|\vec{b}|$, $\vec{a} + \vec{b}$ and $2|\vec{a} + 5\vec{b}|$. [1+1+1+2] (b) Estimate the value of $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 9} - 3}{x^2}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. (a) The area of the parabola $y = x^2$ from $(1,1)$ to $(2,4)$ is rotated about the y -axis. Find the area of the resulting surface. (b) Find the solution of the equation $y^2 dy = x^2 dx$ that satisfies the initial condition $y(0) = 2$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. As dry air moves upward, it expands and cools. If the ground temperature is (20°C) and the temperature at height of 1 km is (10°C) , express the temperature T (in $^\circ\text{C}$) as a function of the height h (in kilometer), assuming that linear model is appropriate. (a) Draw a graph of the function in part (a). What does the slope represent? (c) What is the temperature at a height of (2.5 km) ?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Integrate $\int_0^1 x^2 \sqrt{x^3 + 1} \, dx$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Find the Maclaurin series expansion of $f(x) = e^x$ at $x = 0$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find $\frac{dy}{dx}$ if $x^3 + y^3 = 6xy$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Show that $y = x - \frac{1}{x}$ is a solution of the differential equation $xy' + y = 2x$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Sketch the graph and find the domain and range of the function $f(x) = 2x - 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Determine whether the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2 + 4}$ converges or diverges.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. If $f(x,y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2,1)$ and $f_y(2,1)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Show that the function $f(x) = x^2 + \sqrt{7 - x}$ is continuous at $x = 4$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.